What Survives the Spread?

A pre-registered evaluation of popular retail trading methods
on CFDs, futures and crypto order books, 2021–2026

Davide Maccabelli
Independent research — built with the *NullEdge* research and execution framework

October 2026 — working paper

Abstract

Retail traders are taught many discretionary methods: liquidity and time-cycle models in the ICT/SMC tradition, order-book "walls", multi-module indicators, opening-range breakouts and classic calendar anomalies. Most retail accounts lose money. I turn these methods into explicit rules and test them with a protocol built to avoid self-deception. For each study, the hypotheses, parameters and pass criteria are written down before the data are examined. Each study is run once on bid/ask-aware execution with the costs of a real broker. Results are compared with random-entry baselines and split into separate development and verification periods. Across 19 pre-registered studies and about 50 primary tests on the Nasdaq-100, S&P 500, Dow, DAX, gold and Bitcoin, **no intraday method produces an expectancy distinguishable from zero, let alone positive, net of retail CFD costs**. Many have a small gross edge of $+0.01$ to $+0.09$ R per trade, where R is the initial risk, while costs are 0.06 to 0.37 R per trade. Order-book signals recorded live from 14 crypto exchange feeds are statistically detectable but five to twenty times smaller than the spread. The effects that survive all sit at the daily horizon in US equity indices: Connors' RSI(2) pull-back and the internal-bar-strength reversal. With futures-level costs, a 15-minute opening-range breakout on the Nasdaq-100 becomes marginally positive ($+0.044$ R per trade, $t = 1.85$, on micro futures). At 09:45 New York time, the size of the rest of the session is predictable out of sample (AUC 0.74), but whether it will trend, or in which direction, is not (AUC 0.46–0.51). For a retail trader, the instrument and its cost structure matter more than the choice of strategy.

1 Introduction

Regulators report that 74–89% of retail CFD accounts lose money (ESMA, 2018). Among Taiwanese day traders, more than 80% lose after costs and only a few are persistently profitable (Barber et al., 2014). Among Brazilians who day-traded index futures for more than 300 days, 97% lost money (Chague et al., 2019). Retail education nevertheless promotes many intraday methods, usually taught as discretionary frameworks illustrated on hand-picked charts.

In this paper I ask a narrow question: *when these methods are written down as explicit rules and tested honestly, with real costs, do any of them have an edge?* A second question follows: *if not, what does survive, and what would it take to trade it?*

The contribution is mostly methodological and empirical:

1. A research and execution framework (*NullEdge*) that runs the *same* rule code in backtest, in a "shadow" mode on live prices and in live trading. It enforces pre-registration, random baselines and an off $\rightarrow$ shadow $\rightarrow$ live promotion ladder with automatic demotion.
2. Results of 19 pre-registered studies, reported in full, including all negative results and every variant tried.
3. A decomposition of gross edge versus cost, showing that cost, not signal, is the binding constraint for intraday trading on retail CFDs.

4. An out-of-sample test of whether trend days can be recognised at the open.
5. Evidence from a live deployment on a demo account, including the cost of running an unvalidated strategy.

2 Related literature

Backtest overfitting and multiple testing are well documented. Bailey et al. (2017) formalise the probability of backtest overfitting, Bailey and López de Prado (2014) propose the deflated Sharpe ratio, and Harvey et al. (2016) argue for $t > 3$ hurdles. McLean and Pontiff (2016) find that published anomalies earn 26% less out of sample and 58% less after publication. On intraday predictability, Gao et al. (2018) document market intraday momentum and Baltussen et al. (2021) link it to hedging demand. Zarattini and Aziz (2023) and Zarattini et al. (2024) report profitable opening-range and noise-area momentum rules on QQQ and SPY at very low ETF costs. Lou et al. (2019) show that reversal and momentum profits accrue overnight. Connors and Alvarez (2008) popularised the RSI(2) pull-back. At the microstructure level, order-flow imbalance explains *contemporaneous* price changes over 10-second intervals (Cont et al., 2014), and queue imbalance predicts the *next tick* (Gould and Bonart, 2016). Both horizons are far shorter than a retail holding period. For machine learning, Gu et al. (2020) find gains mainly in cross-sectional stock selection at monthly horizons. Fischer and Krauss (2018) report LSTM profits that fade after 2010. Avramov et al. (2023) trace deep-learning profits to hard-to-arbitrage stocks, and Prata et al. (2024) find that order-book deep-learning models degrade sharply on new data. Cycle-based methods go back to Hurst (1970); the only cycle with academic support is lunar (Dichev and Janes, 2003).

3 Data

Dataset	Content	Period	Size
Broker MT5 bars	CFD bars M5–D1 for about 20 symbols; spread per bar on M5–M15	2014–2026 (M5 from 2025)	215 MB
Dukascopy M1	Nasdaq-100, S&P 500 and gold, 1-minute bars (proxy for futures and CFDs)	2021-06 – 2026-10	143 MB
Binance 1m	BTC and ETH spot and perpetual, PAXG spot, with aggressor (taker) volume; funding	2022 – 2026	631 MB
L2 recorder	Book snapshots every 10 s from 14 exchange feeds (40 liquidity bands of 0.05% per side, top-10 orders), aggressor flow; BTC, ETH, PAXG	from 2026-10-03	1.1 GB
Deribit options	BTC and ETH option chain every 5 minutes (OI, mark IV)	from 2026-10-12	accumulating
Daily series	11 equity indices, ETFs, FX, BTC, Fed funds (Yahoo); broker D1 for 35 symbols	2000 – 2026	19 MB
Trade journal	Every live and shadow trade (139), every signal decision (21,121), equity every 5 minutes	2026-09-29 –	15 MB

Table 1: Data used. In total about 2.3 GB, which I collected and maintain.

4 Methodology

4.1 Pre-registration

Each study starts with a written document that fixes the hypotheses, the exact rules and parameters, the data split, the costs and the pass criteria, before any result is seen. The study is then run *once*, and every variant is reported, including those that fail. A few pre-registrations were written in the same working session as the code. They are an honest list of everything that was tried, not an independent commitment, and I say so where it applies.

4.2 Execution model

Decisions are made at a bar close using only information available at that time; higher time frames are aligned on their closing time. Execution follows MetaTrader 5 semantics:

- bars are on the bid, and longs are filled at the ask;
- the spread is the larger of the bar spread and the 90th percentile spread for that hour;
- limit orders fill only if the price trades through by one tick;
- when a bar touches both stop and target, the stop counts;
- a gap through the stop exits at the open.

For crypto, market orders also pay half a spread of slippage and stop orders a full spread. Multi-day positions pay the broker's swap or CFD financing (Fed funds $+2.5\%$ for longs).

4.3 Costs

Three cost regimes are compared for the Nasdaq-100: the broker CFD (3.0 points round trip), Micro E-mini futures (about 1.0 point including commission, \$2 per point) and E-mini futures (about 0.5 point). The other CFD costs are 3.9 bps for the S&P 500, \$0.30 for gold and 6.25 bps for BTC.

4.4 Baselines and validation

Every strategy is compared with a baseline that has the same exits: entries in the same bars with a random direction, or entries at random times. For long-only rules, the baseline is the same exposure at other times of day. A rule has an edge only if it clearly beats its own baseline. The primary criteria are:

- net expectancy positive in both the development and the verification period;
- $t \geq 2.5$ on the full sample (Bonferroni-adjusted in some families, deflated Sharpe in one study);
- still positive with costs $\times 1.5$;
- neighbouring parameter values also positive.

Intraday studies use development 2021-07 to 2024-12 and verification 2025-01 to 2026-10.

4.5 Deployment ladder and risk

A strategy moves from *off* to *shadow* (virtual trades on live prices with exact sizing) to *live* only after passing its study. A monitor demotes a live strategy back to shadow if its last 20 or more trades fall significantly below the backtest ($t < -2$). Portfolio rules:

- 0.25–1% risk per trade, scaled to the strength of evidence;
- at most 1.5% open risk, with per-cluster caps;
- daily (-1.5%) and weekly (-4%) loss limits;
- a spread guard;

- a broker-side stop that is checked every 10 seconds;
- a manual kill switch.

5 Results

I distribute the complete table of results (56 rows) with this paper. Table 2 condenses it by family.

Family (examples)	Tests	Best net result	Verdict
ICT/SMC liquidity models (90-min cycles, purge & revert, SMT, Judas swing, Sunday gap)	12	-0.025 R	none pass
Commercial 90-minute time-cycle method; Hurst cycles; lunar cycle	6	-0.039 R; 0/64	none pass
Proprietary multi-module indicator (God Vision 1 and 2)	3	$+0.002$ R	none pass
Order flow and order book (walls, delta, large orders, funding, PAXG lead)	10	-0.081 R	none pass
Intraday momentum and breakouts (noise area, VBO, LVM, ORB, gap fade, gold Asian session)	13	$+0.007$ R ($t = 0.16$)	none on CFD; ORB on futures
Daily anomalies (RSI(2), IBS, overnight, turn of month, trend following)	9	$+0.210$ R	RSI(2) and IBS on US indices

Table 2: Summary by family. "Best net result" is the highest net expectancy per trade among the primary tests, at the broker's CFD costs.

5.1 Discretionary liquidity and time-cycle frameworks

The ICT "purge and revert" entry model on 90-minute cycles produced 5,331 trades on the Nasdaq-100 and 4,713 on gold. Net expectancy was -0.236 R ($t = -9.7$) and -0.264 R ($t = -11.3$). Gross expectancy was $+0.034$ R and -0.018 R, and targets were hit no more often than on shuffled data (23.9% versus 25.5% by chance). Smart-money divergence (SMT) made results worse, not better. Four further rules from the 2022 mentorship series (morning window, Judas swing at midnight, Sunday gap, 09:30 rebalance) were all negative; the best was the Judas swing on the Nasdaq, at -0.025 R net and $+0.049$ R at futures costs ($t = 0.92$). A commercial time-cycle method (Asian-range sweep recovered in London; fair-value-gap inversion) hit its targets at chance rates. Hurst-style cycles, tested on 16 markets from 2000, showed no more regularity than surrogate series with shuffled return signs. Forward-line (FLD) signals and the cyclical-trough window failed (0/48), and so did the lunar effect (0/16).

5.2 A proprietary multi-module indicator

God Vision, my own indicator, combines Hurst-adaptive channels, volume clustering, an estimated delta, Fisher transforms, a Fokker–Planck projection and liquidity pockets. Rewritten in Python without look-ahead, its trend trigger on gold returned $+0.002$ R over about 3,900 trades. That equals a random long entry with the same exits. Its successor, God Vision 2, did *worse* than random entries on gold (-0.278 R against -0.142 R): the confirmation filter selected the wrong trades. The one module with information, a multi-time-frame bias aligned with the 4-hour trend ($+0.14$ ATR over 4 hours, $t = 3.2$, long side only), did not survive stops and costs.

5.3 Order flow and the order book

Book "walls" (bands holding at least three times the median liquidity, persistent for at least one minute) were reached by price 188 times in the first week. They held 52.1% of the time against 50.5% for control levels at the same distance ($z = 0.24$). A model combining confluences

ranked bounces *worse* than chance (AUC 0.46). Aggressor delta showed a slight 15-minute reversal ($t = -2.2$), but the extreme deciles were worth -0.7 to -3.4 bps against a 12.5 bps cost. Large orders (≥ 30 BTC) were followed by moves *against* them (-19.4 bps, $t = -4.0$), and the effect did not replicate on ETH. Four spot-versus-perpetual and funding rules on BTC from 2022 lost between -0.08 and -0.19 R net, with gross results near zero. The clearest example of "significant but useless" is PAXG leading spot gold: the information coefficient is significant in both periods ($t = 3.7$ and 3.4), but the economic value is 0.07–0.48 bps, far below the cost.

5.4 Intraday momentum, breakouts and the role of costs

Figure 1 sets gross against net expectancy for the intraday rules whose cost decomposition is available. Every gross edge above zero sits between $+0.01$ and $+0.09$ R; costs are between 0.06 and 0.37 R per trade. The only intraday rule with a positive net result on CFDs, a long position in gold from the Asian reopening to the London open, made $+0.007$ R per trade ($t = 0.16$), which is indistinguishable from zero. Two cases deserve attention. The low-volatility 60-minute momentum rule had a robust gross edge in development ($+0.147$ R, $t = 3.9$) that disappeared in verification (-0.033 R gross), even at micro-futures costs. Meta-labeling with purged cross-validation failed its pre-registered gate, so the verification period was never touched.

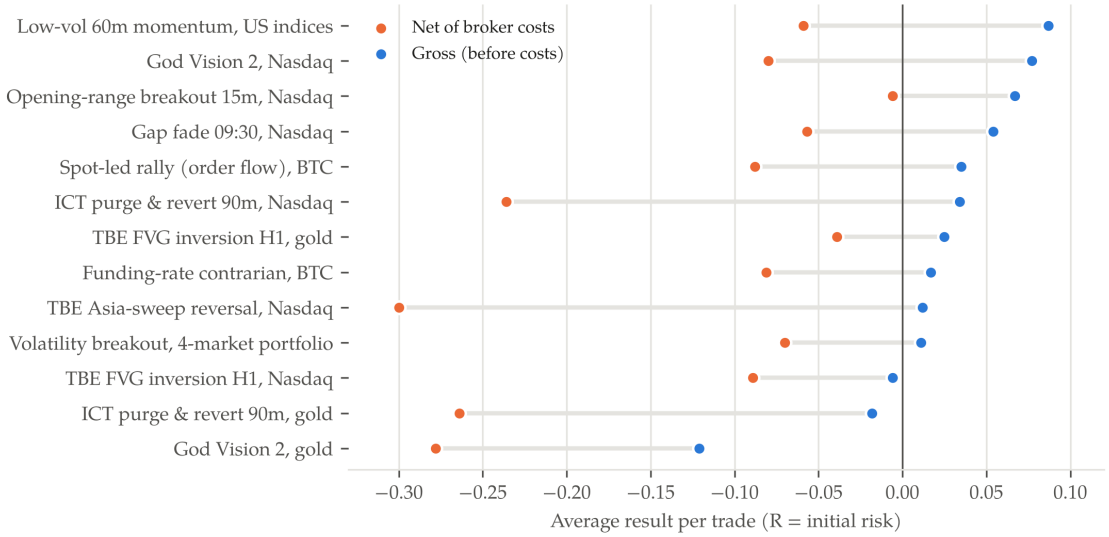


Figure 1: Gross (blue) versus net (orange) expectancy per trade for intraday rules. Costs are those of the retail broker used in each study.

The 15-minute opening-range breakout (ORB) on the Nasdaq-100 enters on a 5-minute close beyond the range, with a stop on the opposite side, a 2R target and an exit at 16:00. It makes 1.56 trades per session and has a stable gross edge of $+0.067$ R. Figure 2 shows that its sign depends entirely on cost:

- at CFD cost: -0.006 R ($t = -0.26$);
- on micro futures: $+0.044$ R ($t = 1.85$; development $+0.047$, verification $+0.038$);
- on E-mini futures: $+0.057$ R ($t = 2.39$).

The edge comes from trend days held to the close, not from the targets, which are hit *less* often than by chance. By year, on micro futures, the strategy made $+16$ R in 2022, $+8$ R in 2023, $+12$ R in 2024, $+24$ R in 2025 and $+1$ R in 2026 to date. With fractional sizing at 1% risk this would have compounded at about 17% a year, with a -27% maximum drawdown. That figure is hypothetical. Since 2025 the median stop is 126 points, or \$252 per micro contract, so 1% risk with one contract requires an account of about \$25,000. On the \$7,700 demo account only 6% of signals would have been tradable.

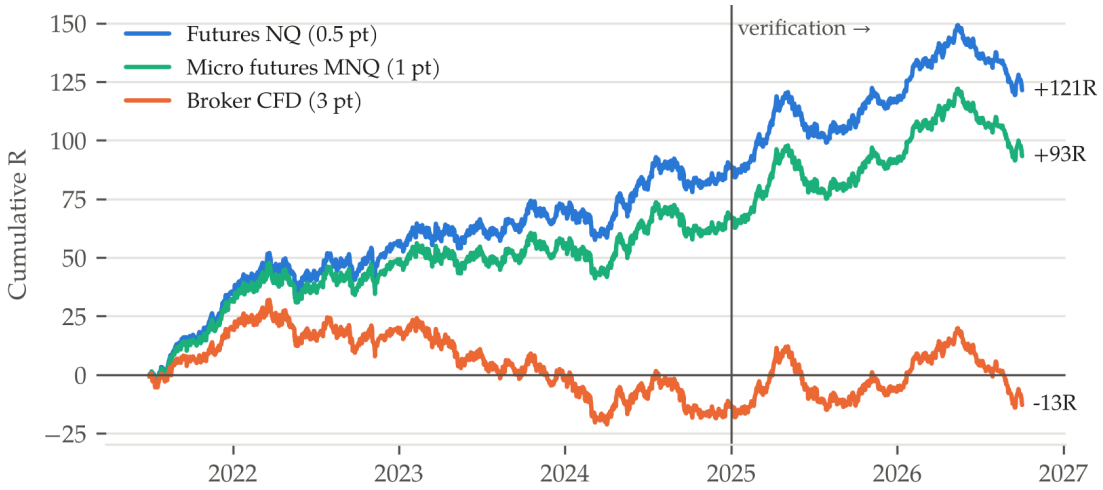


Figure 2: Cumulative R of the 15-minute ORB on the Nasdaq-100 under three cost regimes. Same signals, different cost.

5.5 What survives: daily anomalies in US indices

Two effects pass:

- **Connors' RSI(2) pull-back above the 200-day average,** decided on the hourly bar that closes at 16:00 New York time. It earned +0.210 R per trade on the S&P 500 CFD ($t = 3.79$) and +0.151 R on the Nasdaq CFD ($t = 3.01$) over 2020–2026. That is out of sample with respect to the 1995–2007 period in which it was popularised. On the Dow it earned +27.6 bps per trade ($t = 2.83$); outside the US it failed.
- **The internal-bar-strength (IBS) reversal on the Nasdaq-100:** +11.2 bps per day in position, $t = 4.14$, 2000–2026, positive in both halves. The executable version since 2021, using an intraday proxy at 15:59 and CFD costs, is much weaker (+2.95 bps, $t = 0.57$).

Both effects are specific to US indices: the same IBS rule is negative on seven of eight non-US indices, on metals and on oil, and flat on 31 currency pairs. This is consistent with reversal profits accruing overnight (Lou et al., 2019). The US overnight drift itself is positive but weak (+0.022 R per night, $t = 1.86$). Turn-of-the-month (0/11), pre-holiday (0/3) and multi-market trend following (Sharpe 0.15) do not pass. These edges are real but small and infrequent: about 10 trades a year per index for RSI(2).

5.6 Can trend days be recognised at the open?

Discretionary traders often say the hard part is knowing whether the day will trend or chop. At 09:45 New York time I measured three targets for the rest of the Nasdaq session: its range relative to ATR, its trend efficiency ($|\text{close} - \text{open}|/\text{range}$) and its direction. The predictors are the overnight range, the gap, the opening-range size and direction, yesterday's range and efficiency, NR7, the VIX and FOMC days. Models were fitted on 2021–2024 and evaluated once on 2025–2026.

Volatility is predictable: the logistic AUC for a wide day is 0.74, driven by the overnight and opening ranges. Trend efficiency and direction are not (AUC 0.46–0.51). Using predicted trendiness to filter the ORB did not help in verification (−0.017 R, $t = -0.20$). The same family of tests shows that a session's trendiness does not carry over from the previous days (Spearman −0.10 to +0.04 on four markets).

5.7 Live deployment

The framework has traded a $8,930 demo account since 29 September 2026 (Figure 4). An order-book scalper was run live *before* its research was complete: 81 trades, 17% winners, −35 R,

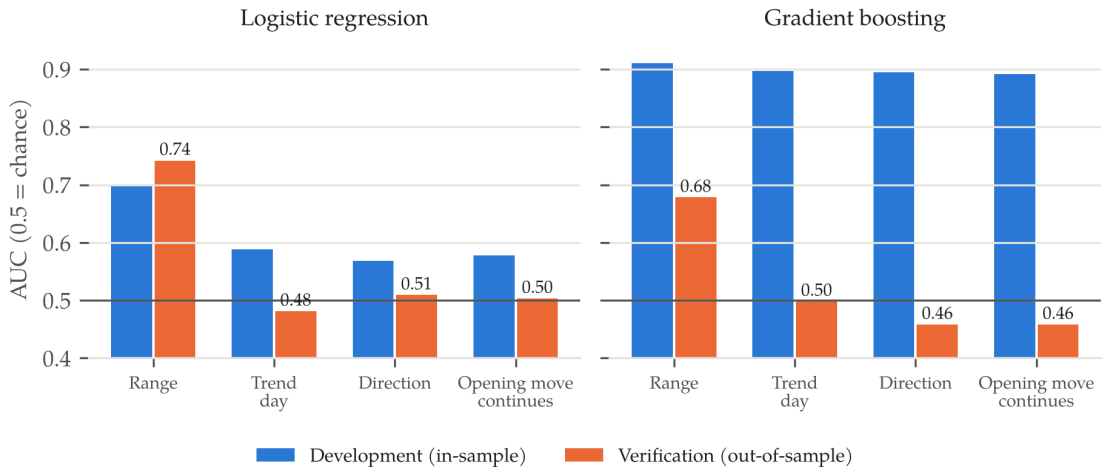


Figure 3: Out-of-sample AUC for four targets at 09:45. Range is predictable; trend and direction are not. Gradient boosting fits the development period almost perfectly (AUC ≈ 0.90) and falls to chance in verification: overfitting in its plainest form.

or about $-1{,}400$ USD in three days. Its later study found its signals below the spread at every horizon. Since 2 October the account has traded only validated strategies, plus two small live tests at 0.5% risk that were labelled as such, and equity has moved -1.1% in 11 days on a handful of trades. That is too few to judge in either direction.

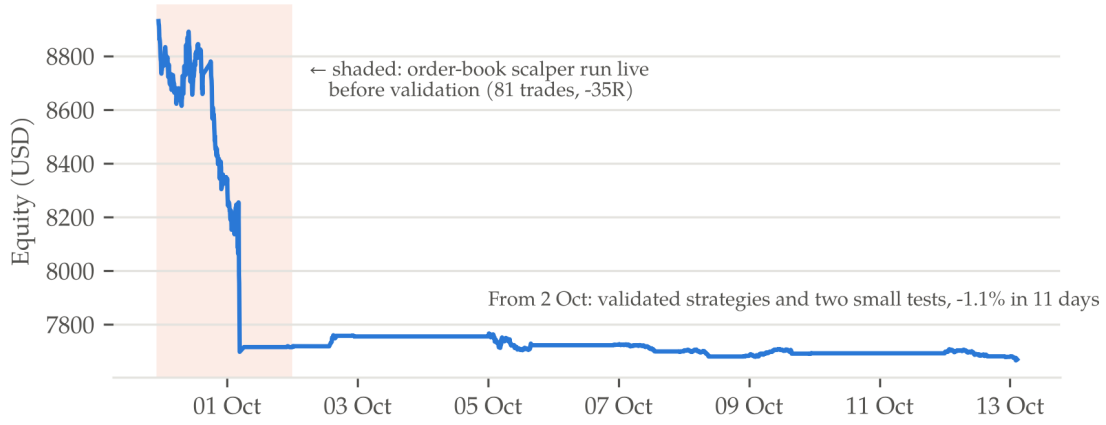


Figure 4: Equity of the demo account. The shaded period is the unvalidated scalper.

6 Discussion

Cost decides. The consistent pattern is a small gross edge consumed by a larger cost. Popular intraday methods are not obviously worse than random before costs; they are simply not better by enough. The one candidate that turns positive does so only when the cost per trade falls from 3 to 1 point. For a retail trader, the first decision is the instrument and its cost structure, and the choice of strategy comes after.

Significant is not profitable. Microstructure signals are real. The PAXG lead and the aggressor-delta reversal are statistically significant, but their size is a fraction of a basis point to a few basis points. This matches the literature: order-flow imbalance explains prices over seconds (Cont et al., 2014; Gould and Bonart, 2016), a horizon at which a retail spread is prohibitive.

Frequency is not an edge. The most frequent strategies in this study were also the most negative. The order-book scalper made 81 trades in three days at -0.44 R per trade. More trades compound the sign of the expectancy; they cannot change it.

Machine learning. Flexible models fit the development period almost perfectly and fail out of sample (Figure 3), as the literature on order-book deep learning also reports (Prata et al., 2024). Where machine learning has documented value, in cross-sectional selection (Gu et al., 2020) and volatility forecasting, its use is different from timing a single index.

Limitations.

- The intraday sample covers 2021–2026, a single macro regime with high rates.
- Dukascopy data is a proxy for futures and CFDs. Where both exist, the same ICT test is negative on both over the same three months (-0.175 R on the proxy, -0.084 R on the broker CFD).
- Broker index spreads rose two- to threefold in June 2026, so some CFD results reflect current, not historical, costs.
- The L2 tests cover only the first one to two weeks of recording.
- Discretionary methods were translated into rules from public material. A skilled discretionary trader may apply judgement that the rules do not capture, so the results speak to the *rules as written*.
- About 50 primary tests were run across families. At a 5% level, two or three false positives would be expected by chance, which is why the hurdles were higher than $t = 2$.
- Three studies (spot-versus-perpetual flow, God Vision as a filter, PAXG) did not record their verdicts in the pre-registration document. Their numbers here are recomputed from the logged trades.

7 Conclusion and future work

After costs, none of the popular intraday methods tested has an edge on retail CFDs. What survives is a small set of daily effects on US equity indices and, with futures-level costs, one intraday breakout. Work planned with the same protocol:

- a joint futures portfolio of the ORB and the daily effects, analysed for correlation and combined drawdown;
- volatility-scaled sizing based on the predictable part of intraday range;
- a pre-registered study of Bitcoin dealer-gamma regimes from the Deribit option chain, with the first analysis on 9 November 2026;
- verification of the one post-hoc order-flow hypothesis on unseen days.

Disclaimer. This is independent research on historical and demo-account data. It is not investment advice. Past results, simulated or live, do not guarantee future results.

References

Avramov, D., Cheng, S., Metzker, L. (2023). Machine learning vs. economic restrictions: evidence from stock return predictability. *Management Science* 69(5).

Bailey, D. H., López de Prado, M. (2014). The deflated Sharpe ratio. *Journal of Portfolio Management* 40(5).

Bailey, D. H., Borwein, J., López de Prado, M., Zhu, Q. J. (2017). The probability of backtest overfitting. *Journal of Computational Finance* 20(4).

Baltussen, G., Da, Z., Lammers, S., Martens, M. (2021). Hedging demand and market intraday momentum. *Journal of Financial Economics* 142(1).

Barber, B. M., Lee, Y.-T., Liu, Y.-J., Odean, T. (2014). The cross-section of speculator skill: evidence from day trading. *Journal of Financial Markets* 18.

Chague, F., De-Losso, R., Giovannetti, B. (2019). Day trading for a living? Working paper, SSRN 3423101.

Connors, L., Alvarez, C. (2008). *Short Term Trading Strategies That Work*. TradingMarkets.

Cont, R., Kukanov, A., Stoikov, S. (2014). The price impact of order book events. *Journal of Financial Econometrics* 12(1), 47–88.

Dichev, I. D., Janes, T. D. (2003). Lunar cycle effects in stock returns. *Journal of Private Equity* 6(4).

European Securities and Markets Authority (2018). ESMA agrees to prohibit binary options and restrict CFDs to protect retail investors. Press release, 27 March 2018.

Fischer, T., Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research* 270(2).

Gao, L., Han, Y., Li, S. Z., Zhou, G. (2018). Market intraday momentum. *Journal of Financial Economics* 129(2).

Gould, M. D., Bonart, J. (2016). Queue imbalance as a one-tick-ahead price predictor in a limit order book. *Market Microstructure and Liquidity* 2(2).

Gu, S., Kelly, B., Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies* 33(5).

Harvey, C. R., Liu, Y., Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies* 29(1).

Hurst, J. M. (1970). *The Profit Magic of Stock Transaction Timing*. Prentice-Hall.

Lou, D., Polk, C., Skouras, S. (2019). A tug of war: overnight versus intraday expected returns. *Journal of Financial Economics* 134(1).

McLean, R. D., Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance* 71(1).

Prata, M., Masi, G., Berti, L., et al. (2024). LOB-based deep learning models for stock price trend prediction: a benchmark study. *Artificial Intelligence Review* 57.

Zarattini, C., Aziz, A. (2023). Can day trading really be profitable? Working paper, SSRN 4416622.

Zarattini, C., Barbon, A., Aziz, A. (2024). Beat the market: an effective intraday momentum strategy for S&P500 ETF (SPY). Swiss Finance Institute Research Paper 24-97.